ABIRA WISNUNGGAL

AI ENGINEER

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https://CodeByAbi.github.io/portfolio-abi/

SUMMARY

Abira is a 6th semester Data Science undergraduate at Nusa Mandiri University with strong expertise in Python, SQL, Deep Learning, MLOps, and AI systems such as RAG and LLM-based applications. He has built end-to-end projects including rainfall prediction using XGBoost (82.4% accuracy), object detection with YOLOv8 (91.2% mAP), and AI DemoLabs systems such as PDF-based QnA, RAG pipelines, Text-to-SQL engines, and automated chart generation from natural language. He is experienced in working with PostgreSQL, MySQL, Firebase, and Looker Studio for data management and visualization, and has demonstrated leadership and communication skills as Head of Public Relations and Curriculum Developer, leading initiatives and mentoring peers to improve data science literacy.

EDUCATION

Nusa Mandiri University - **Depok, Indonesia** **Sep 2023 - Present**

Data Science: Current GPA (3.67 / 4.00)

- Engaged in projects focusing on machine learning, data mining, computer vision and natural language processing
- Completed several applied projects including analytics dashboards and machine learning model development using real-world datasets from the Central Statistics Agency (BPS).
- Relevant courses: Machine Learning, Data Visualization, Statistical Inference, Natural Language Processing, Cloud, Text Mining, Deep Learning and Data Mining.
- Honor & Achievements: Top 5 Abstract Competition, Bronze Medal PeDaS x APTIKOM (Pesta Data Nasional) 2025, Third International Contest of Internet of Things & Qualified Scientific Paper organized by Bank Indonesia.

PROFESSIONAL EXPERIENCE

Nawa Darsana Technology (Nawatech) - **Bintaro, Tangerang** **Okt 2025 - Feb 2026**

Machine Learning Engineer Intern

Nawa Darsana Technology is an Indonesia-based technology company focused on developing AI-driven and data-centric solutions for businesses.

- Developed asynchronous AI pipeline using RabbitMQ & Celery with Docker based microservices architecture.
- Engineered Text-to-SQL system with pgvector integration, embedding database schemas and query examples, supporting 3 chatbot capabilities (QnA, PDF-RAG, Text-to-SQL).
- Improved chatbot accuracy & reliability by implementing 4 evaluation metrics (faithfulness, relevancy, hallucination detection, contextual precision/recall) and enhancing validation for both text-based and table-based outputs.

GAO TEK INC. - **New York, America** **Aug 2025 - Jan 2026**

AI Chatbot Development Internship (Remote)

A global technology company that provides various solutions for industries such as IoT, networking, testing and measurement, and drones.

- Conducted applied research on structured data modeling and large language models, implementing experimental improvements to enhance system performance and robustness.
- Designed scalable AI pipelines using Python, PostgreSQL, and containerized environments, integrating third-party tools and automating data processing workflows.
- Performed debugging, performance profiling, and system upgrades while maintaining clean, modular, and well-documented code to support production-ready AI solutions.

HOME CREDIT INDONESIA - Kebagusan, Jakarta **Sep 2025 - Nov 2025**

Data Scientist Intern (Remote)

Home Credit Indonesia is a technology-driven consumer finance company that provides point of sale financing solutions for customers across Indonesia.

- Assisted in data collection, cleaning, and preprocessing using Python and SQL to ensure accuracy and consistency in analytical workflows.
- Utilized Excel for data summarization, reporting, and performance tracking to support day-to-day data operations.
- Created data visualizations and dashboards to communicate findings effectively to the analytics team and stakeholders.

PPM AL FAQIH MANDIRI - Depok, Indonesia **Oct 2023 - Mei 2025**

Data Analyst

A division responsible for determining the curriculum and developing an effective and ideal teaching system at the foundation.

- Designed 3 performance dashboards in Google Data Studio
- Developed interactive data visualizations using Google Data Studio to track syllabus coverage and
- assessment outcomes across departments.
- Automated performance reporting in Google Data Studio.
- Collaborated with cross-functional teams to ensure curriculum alignment with learning objectives.

PROJECTS

Demolabs AI Chatbot **Okt 2025 - Feb 2026**

- Built an end-to-end AI-powered analytics assistant integrating RAG-based Text-to-SQL generation and chart visualization. Designed semantic intent detection and post query data shape parsing to enable dynamic visualization routing, including implicit trend and ranking recognition.
- Implemented async task processing with Celery workers, Redis-based session memory, Azure OpenAI embeddings for similarity search, and vector-based example retrieval to improve SQL accuracy and response relevance.

Smart FAQ Chatbot **Sept 2025 - Okt 2025**

- Deployed via Streamlit and Docker, with Kubernetes support for scaling.
- Achieved an average answer quality score of 0.85 (cosine similarity + LLM evaluation), enabling reliable and secure automated customer support.

Sentiment Analysis Telco Tweets **Jul 2025 - Aug 2025**

- Conducted sentiment analysis on telecom customer tweets using NLP and machine learning (TF-IDF + Logistic Regression).
- Delivered 0.82 F1-score and identified top negative keywords, enabling data-driven strategies to improve telecom service quality.

Time Series Forecasting Top 5 Car Sales Indonesia **Jun 2025 - Sep 2025**

- Implemented end-to-end workflow data preprocessing, trend and seasonality analysis, and forecasting using SARIMA, Prophet, and LSTM models.
- Achieved competitive accuracy with SMAPE below 10% across brands, providing reliable forecasts for September 2025.

Rainfall Prediction in Australia using XGBoost & Random Forest **Jul 2024 - Sep 2024**

- Built a classification model using real-world weather data from the Australian Bureau of Meteorology.
- Achieved accuracy of 82.4%, with precision of 81% and recall of 79%, supporting early warning systems and risk analysis.

ORGANIZATIONAL & COMMITEE EXPERIENCE

HIMPUNAN MAHASISWA SAINS DATA UNM (HIMASATA) - Depok, Indonesia **Jul 2023 - Jan 2025**

Head of Public Relation Division & Head of Information & Communication Division

An organization responsible for uniting students, fostering academic and professional development, and promoting initiatives related to data science at Nusa Mandiri University.

- Published 12 data science articles on Kompasiana to increase public literacy on machine learning.
- Created 12 original data science articles tailored to student needs, covering topics like machine learning fundamentals with an average reach of 30+ readers per article on LinkedIn.

IEEE NUSA MANDIRI UNIVERSITY - Depok, Indonesia **Jul 2024 - Sep 2024**

Public Relation Cordinator & Master of Ceremony

An official student organization at Universitas Nusa Mandiri committed to fostering excellence in engineering, technology, and scientific research, aligned with the vision and mission of the global IEEE community.

- Designed and executed campus-wide public relations strategy aligned with IEEE brand guidelines, reaching over 100 students.
- Managed media coverage and internal communication to increase event visibility by 30% across digital platforms.

PAGE 8.0 (PARENT GATHERING) - Depok, Indonesia **Jul 2024 - Sep 2024**

Project Officer (PO)

A student report distribution program aimed at strengthening the relationship between school and parents at PPM Al Faqih Mandiri, attended by over 100 students from various universities in Depok.

- Led a cross-functional committee of 35 members to execute a student-parent engagement event attended by 100+ participants from 10+ universities.
- Implemented risk mitigation strategies that reduced last-minute disruptions.

CERTIFICATIONS

Associate Data Scientist - BNSP
Python Advance for Data Science - Skilvul
Data Analytics Intensive Bootcamp - Revou Bootcamp
Data Visualization - Dicoding Indonesia

ADDITIONAL INFORMATION

- **Languages:** Bahasa Indonesia (Native), English (Professional Working Proficiency).
- **Technical Skills:** Python, R Programming, SQL, Machine Learning, Deep Learning, Spreadsheets, Excel, Data Visualization, Tableau, Google Data Studio, RabbitMQ, ArgoCD, Statistics.
- **Soft Skills:** Problem Solving, Critical Thinking, Communication, Fast Learner, Pro Active, Collaboration, Team Work.
- **Honor & Awards:** Top 5 Abstract Competition, Bronze Medal PeDaS x APTIKOM (Pesta Data Nasional) 2025, Third International Contest of Internet of Things & Qualified Scientific Paper organized by Bank Indonesia.